

Midterm Review

Suggested Study Plan

1. Understand all concepts and theorems from the cheatsheet.
2. Review all HW problems
3. Solve the midterm review problems
4. (If Needed) Extra Practice from VMLS book

Plan for today

1. Resources and study strategy
2. Problem Solving
 - a. Linear Algebra Review
 - b. K-Means
 - c. KNN
 - d. Least Squares
3. Q&A Session

Plan for today

1. **Resources and study strategy**

2. Problem Solving

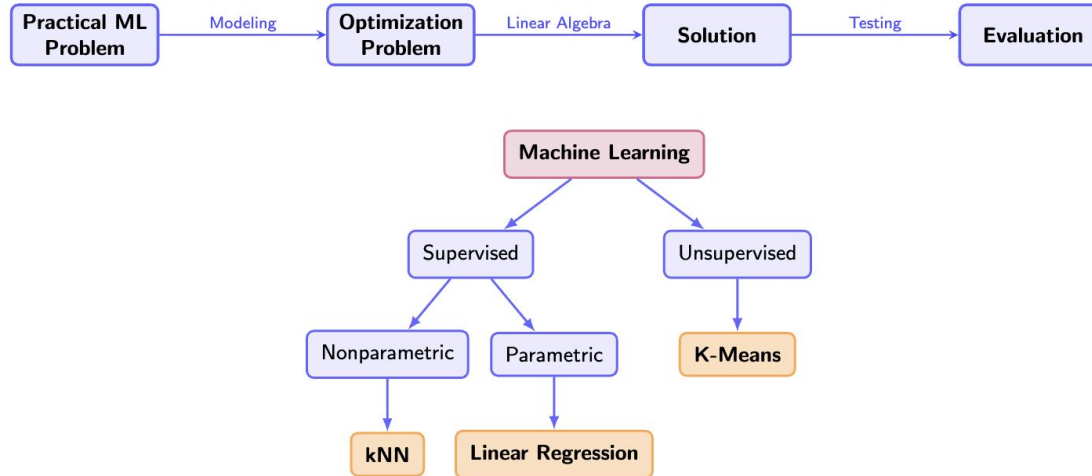
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Review Cheatsheet

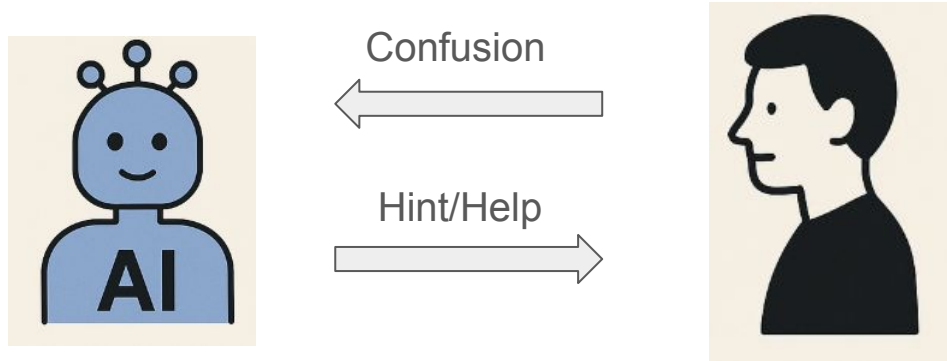
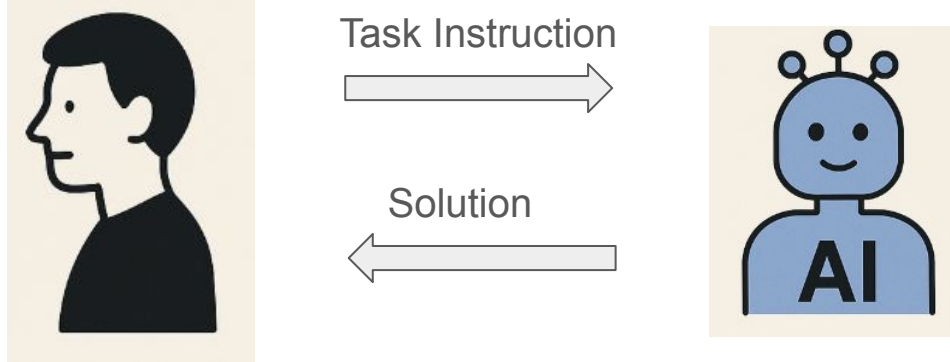
Midterm Reference & Toolbox

This document contains key definitions, formulas, and algorithms for the midterm exam.



Quick Navigation: 1. Vectors and Matrices (p.2) 2. Vector Spaces and Eigenvalues (p.4) 3. K-Means Clustering (p.7) 4. K-Nearest Neighbors (p.8) 5. Least Squares (p.9)

chatGPT study mode



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Pólya's Problem-Solving Method



Problem-solving is iterative: return to earlier steps as needed

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Problem 1

Problem 1. (★ VMLS 1.10 Total score.) The record for each student in a class is given as a 10-vector r , where $r_1, \dots, r_8$ are the grades for the 8 homework assignments, each on a 0 – 10 scale, r_9 is the midterm exam grade on a 0 – 120 scale, and r_{10} is the final exam score on a 0 – 160 scale. The student's total course score s , on a 0 – 100 scale, is based 25% on the homework, 35% on the midterm exam, and 40% on the final exam. Express s in the form $s = w^T r$. You can give the coefficients of w to 4 digits after the decimal point.

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Strategy: First write s as a linear combination of entries in r . Then, fit it into the inner product form to extract w .

Problem 1 - Solution

Step 1: Scale each component to 0-100.

- Total Homework Score (scaled): Since each HW is worth 10 points and there are 8 HWs, the total raw HW score is out of 80. To scale to 0-100:

$$\text{Total_HW_Score} = \frac{100}{80}(r_1 + r_2 + \cdots + r_8) = 1.25(r_1 + r_2 + \cdots + r_8)$$

- Midterm Score (scaled): The midterm is out of 120, so on a 0-100 scale:

$$\text{Midterm_Score} = \frac{100}{120}r_9 = 0.8333r_9$$

- Final Exam Score (scaled): The final is out of 160, so on a 0-100 scale:

$$\text{Final_Score} = \frac{100}{160}r_{10} = 0.625r_{10}$$

Step 2: Apply the weights to get the total score.

The total course score s is a weighted combination: 25% homework, 35% midterm, and 40% final.

$$\begin{aligned}s &= 0.25 \cdot \text{Total_HW_Score} + 0.35 \cdot \text{Midterm_Score} + 0.40 \cdot \text{Final_Score} \\&= 0.25 \cdot 1.25(r_1 + \cdots + r_8) + 0.35 \cdot 0.8333r_9 + 0.40 \cdot 0.625r_{10} \\&= 0.3125(r_1 + \cdots + r_8) + 0.2917r_9 + 0.25r_{10}\end{aligned}$$

Step 3: Express as an inner product $s = w^T r$.

Writing this as an inner product:

$$s = (0.3125, 0.3125, \dots, 0.3125, 0.2917, 0.25)^T r$$

where the first 8 entries are all 0.3125 (one for each homework).

Therefore, $w = (0.3125\mathbf{1}_8, 0.2917, 0.2500)$, where $\mathbf{1}_8$ denotes an 8-dimensional vector of ones.

Problem 3. (★★ VMLS 3.5 Other common norms.) A vector norm is any real-valued function f satisfying the following four properties:

1. **Nonnegative homogeneity:** $\|\alpha x\| = |\alpha|\|x\|$ for all scalars α and vectors x .
2. **Triangle inequality:** $\|x + y\| \leq \|x\| + \|y\|$ for all vectors x and y .
3. **Nonnegativity:** $\|x\| \geq 0$ for all vectors x .
4. **Definiteness:** $\|x\| = 0$ if and only if $x = 0$.

The 1-norm and ∞ -norm for n -vectors are defined as

$$\|x\|_1 = |x_1| + \cdots + |x_n|, \quad \|x\|_\infty = \max \{|x_1|, \dots, |x_n|\}.$$

Verify that the 1-norm and the ∞ -norm satisfy the four norm properties listed above.

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Verify that the 1-norm and the ∞ -norm satisfy the four norm properties listed above.

Strategy: This is a question of definitions, all of which are provided in the question.

So, we can use the definition of the 1-norm and check if the 4 conditions hold.

a) Homogeneity.

$$\begin{aligned}\|\beta x\|_1 &= |\beta x_1| + |\beta x_2| + \cdots + |\beta x_n| \\ &= |\beta| |x_1| + |\beta| |x_2| + \cdots + |\beta| |x_n| \\ &= |\beta| (|x_1| + |x_2| + \cdots + |x_n|) \\ &= |\beta| \|x\|_1\end{aligned}$$

(b) Triangle inequality.

$$\begin{aligned}\|x + y\|_1 &= |x_1 + y_1| + |x_2 + y_2| + \cdots + |x_n + y_n| \\ &\leq |x_1| + |y_1| + |x_2| + |y_2| + \cdots + |x_n| + |y_n| \\ &= \|x\|_1 + \|y\|_1\end{aligned}$$

(c) Nonnegativity. Each term in $\|x\|_1 = |x_1| + \cdots + |x_n|$ is nonnegative.

(d) Definiteness. $\|x\|_1 = 0$ only if $|x_1| = \cdots = |x_n| = 0$.

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Problem 4. (★★★ Minimizing mean square distance to a set of vectors.) Let $x_1, \dots, x_L$ be a collection of n -vectors. Show that the vector z which minimizes the sum-square distance

$$J(z) = \|x_1 - z\|^2 + \cdots + \|x_L - z\|^2$$

is the centroid $\bar{x} = (1/L)(x_1 + \cdots + x_L)$.

(a) Explain why, for any z , we have

$$J(z) = \sum_{i=1}^L \|x_i - \bar{x} - (z - \bar{x})\|^2 = \sum_{i=1}^L (\|x_i - \bar{x}\|^2 - 2(x_i - \bar{x})^T(z - \bar{x})) + L\|z - \bar{x}\|^2.$$

(b) Explain why $\sum_{i=1}^L (x_i - \bar{x})^T(z - \bar{x}) = 0$. *Hint.* Write the left-hand side as

$$\left(\sum_{i=1}^L (x_i - \bar{x}) \right)^T (z - \bar{x}),$$

and argue that the left-hand vector is 0.

(c) Combine the results of (a) and (b) to get $J(z) = \sum_{i=1}^L \|x_i - \bar{x}\|^2 + L\|z - \bar{x}\|^2$. Explain why for any $z \neq \bar{x}$, we have $J(z) > J(\bar{x})$. This shows that the choice $z = \bar{x}$ minimizes $J(z)$.

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Solution. (a) Using the identity $\|a - b\|^2 = \|a\|^2 - 2a^T b + \|b\|^2$, we rewrite $x_i - z = (x_i - \bar{x}) - (z - \bar{x})$ and expand:

$$\|x_i - z\|^2 = \|x_i - \bar{x}\|^2 - 2(x_i - \bar{x})^T(z - \bar{x}) + \|z - \bar{x}\|^2.$$

Summing over $i = 1, \dots, L$:

$$J(z) = \sum_{i=1}^L \|x_i - \bar{x}\|^2 - 2 \sum_{i=1}^L (x_i - \bar{x})^T(z - \bar{x}) + L\|z - \bar{x}\|^2.$$

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and argue that the left-hand vector is 0.

(b) We can factor out $(z - \bar{x})$ from the middle term:

$$\sum_{i=1}^L (x_i - \bar{x})^T (z - \bar{x}) = \left(\sum_{i=1}^L (x_i - \bar{x}) \right)^T (z - \bar{x}).$$

Since $\bar{x} = \frac{1}{L} \sum_{i=1}^L x_i$, we have $\sum_{i=1}^L (x_i - \bar{x}) = \sum_{i=1}^L x_i - L\bar{x} = 0$. Thus the middle term vanishes.

(c) Combine the results of (a) and (b) to get $J(z) = \sum_{i=1}^L \|x_i - \bar{x}\|^2 + L\|z - \bar{x}\|^2$. Explain why for any $z \neq \bar{x}$, we have $J(z) > J(\bar{x})$. This shows that the choice $z = \bar{x}$ minimizes $J(z)$.

(c) From parts (a) and (b):

$$J(z) = \sum_{i=1}^L \|x_i - \bar{x}\|^2 + L\|z - \bar{x}\|^2.$$

The first term is independent of z , and the second term is nonnegative, achieving its minimum value of 0 when $z = \bar{x}$. Therefore, $\bar{x}$ uniquely minimizes $J(z)$.

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Problem 6. (★★★ Weighted Norm Decision Boundaries.) In weighted 1-NN classification, the decision boundary between two training points $a, b \in \mathbb{R}^d$ is the set of points x satisfying

$$d_W(x, a) = d_W(x, b), \quad \text{where} \quad d_W(x, y) = \sqrt{(x - y)^\top W (x - y)}, \quad W = \text{diag}(\alpha_1, \dots, \alpha_d), \quad \alpha_i > 0.$$

(a) Show that the decision boundary has the form of a hyperplane

$$c^\top x = d,$$

for some vector c and scalar d depending on a, b, W .

(b) Specialize your result to $d = 2$ and write the equation of the decision boundary in terms of $\mathbb{R}^2$ coordinates x_1, x_2 .

(c) Again with $d = 2$, show that if $\alpha_1 \neq \alpha_2$, this boundary is not perpendicular to the segment ab in the original coordinates.

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Strategy: Expand out the decision boundary condition and then rearrange

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(a) Squaring both sides: $(x - a)^\top W (x - a) = (x - b)^\top W (x - b)$. Expanding and canceling $x^\top W x$

$$-2x^\top W a + a^\top W a = -2x^\top W b + b^\top W b \quad \Longleftrightarrow \quad \boxed{(W(b - a))^\top x = \frac{1}{2}(\|b\|_W^2 - \|a\|_W^2)}$$

where $\|u\|_W^2 := u^\top W u$. This is a hyperplane with $c = W(b - a)$ and $d = \frac{1}{2}(\|b\|_W^2 - \|a\|_W^2)$.

b) Specialize your result to $d = 2$ and write the equation of the decision boundary in terms of $\mathbb{R}^2$ coordinates x_1, x_2 .

(b) With $W = \text{diag}(\alpha_1, \alpha_2)$ and $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, substituting into part (a) yields

$$\alpha_1(b_1 - a_1)x_1 + \alpha_2(b_2 - a_2)x_2 = \frac{1}{2} [\alpha_1(b_1^2 - a_1^2) + \alpha_2(b_2^2 - a_2^2)] .$$

Rearranging gives the weighted perpendicular bisector: $(b - a)^\top W \left(x - \frac{a+b}{2} \right) = 0$.

c) Again with $d = 2$, show that if $\alpha_1 \neq \alpha_2$, this boundary is not perpendicular to the segment ab in the original coordinates.

(c) The decision boundary from part (a) is

$$(W(b - a))^{\top} x = \frac{1}{2} (\|b\|_W^2 - \|a\|_W^2),$$

which is a hyperplane with normal vector

$$n = W(b - a).$$

The line segment connecting a and b has direction vector

$$v = b - a.$$

- c) Again with $d = 2$, show that if $\alpha_1 \neq \alpha_2$, this boundary is not perpendicular to the segment ab in the original coordinates.

The boundary is *perpendicular* to the segment ab if and only if the normal n is parallel to v , that is,

$$W(b - a) = \lambda(b - a), \quad \text{for some } \lambda \in \mathbb{R}.$$

This requires that $b - a$ be an eigenvector of W . For

$$W = \text{diag}(\alpha_1, \alpha_2),$$

the eigenvectors are the coordinate axes, so this condition holds only when $\alpha_1 = \alpha_2$ (the isotropic case) or when $b - a$ is aligned with one of the axes. Hence, for generic a, b and $\alpha_1 \neq \alpha_2$, the boundary is *not perpendicular* to the segment ab in the original coordinates.

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Problem 7. (★★ VMLS 12.1 Approximating a vector as a multiple of another one.) Find a scalar x that minimizes $\|ax - b\|^2$, where a and b are nonzero m -vectors. Show that $\|a\hat{x} - b\|^2 = \|b\|^2(\sin \theta)^2$, where $\theta = \angle(a, b)$.

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Strategy:

$$\hat{x} = (a^T a)^{-1} a^T b$$

$$a^T b = \|a\| \|b\| \cos \theta,$$

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The least squares solution is given by

$$\hat{x} = (a^T a)^{-1} a^T b = \frac{a^T b}{a^T a} = \frac{a^T b}{\|a\|^2}.$$

The residual is

$$a\hat{x} - b = a \cdot \frac{a^T b}{\|a\|^2} - b = \frac{a^T b}{\|a\|^2} a - b.$$

The squared norm of the residual is

$$\|a\hat{x} - b\|^2 = \left\| \frac{a^T b}{\|a\|^2} a - b \right\|^2.$$

Problem 7. (★★ VMLS 12.1 Approximating a vector as a multiple of another one.) Find a scalar x that minimizes $\|ax - b\|^2$, where a and b are nonzero m -vectors. Show that $\|a\hat{x} - b\|^2 = \|b\|^2(\sin \theta)^2$, where $\theta = \angle(a, b)$.

To compute this, we expand:

$$\begin{aligned}\|a\hat{x} - b\|^2 &= \left\| \frac{a^T b}{\|a\|^2} a \right\|^2 - 2 \left(\frac{a^T b}{\|a\|^2} a \right)^T b + \|b\|^2 \\ &= \frac{(a^T b)^2}{\|a\|^4} \|a\|^2 - 2 \frac{a^T b}{\|a\|^2} a^T b + \|b\|^2 \\ &= \frac{(a^T b)^2}{\|a\|^2} - 2 \frac{(a^T b)^2}{\|a\|^2} + \|b\|^2 \\ &= \|b\|^2 - \frac{(a^T b)^2}{\|a\|^2}.\end{aligned}$$

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Now, recall that $a^T b = \|a\| \|b\| \cos \theta$, where $\theta = \angle(a, b)$. Substituting:

$$\|a\hat{x} - b\|^2 = \|b\|^2 - \frac{(\|a\| \|b\| \cos \theta)^2}{\|a\|^2} = \|b\|^2 - \|b\|^2 \cos^2 \theta = \|b\|^2 (1 - \cos^2 \theta).$$

Using the trigonometric identity $\sin^2 \theta + \cos^2 \theta = 1$, we get $1 - \cos^2 \theta = \sin^2 \theta$. Therefore,

$$\|a\hat{x} - b\|^2 = \|b\|^2 \sin^2 \theta.$$

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